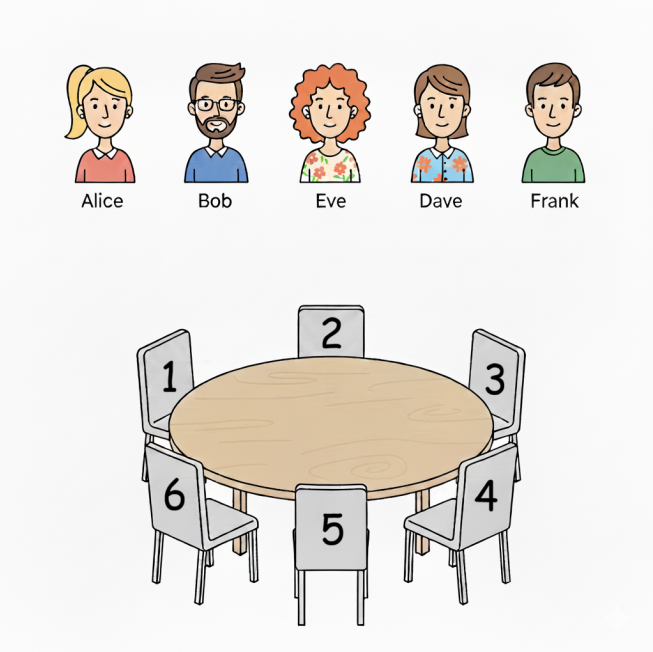
HOMEWORK CSP

Introduction to Artificial Intelligence

Question

There are six people – Alice, Bob, Carol, Dave, Eve, and Frank – and six seats of the circular table indexed from 1-6 (i.e., `seat1` to `seat6`) as shown as the figure below.



The following facts are also known.

- Alice must sit directly opposite Bob.
- Frank refuses to sit next to Alice.
- Dave wants to sit between Eve and Bob.
- Alice wants to sits at `seat1`.

1. Formulate this problem as a CSP. Identify all constraints implied in the problem and represent each of them as a tuple of $\langle scope, relation \rangle$. Note that all higher-order constraints can be formulated as binary constraints.
2. Using your approach (with explanations provided in each step) to give a solution

the puzzle.

Submission Requirements

Write your answers in the following template.

- Student ID: `[Your Student ID]`
- Name: `[Your Name]`
- Answers:
    1. `[Your answer]`
    2. `[Your answer]`

The submission document should be in PDF format and named as `StudentID.pdf`.